

Understand Ratios

Lesson 3-1

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



3 : 2

3 cups flour to 2 cups sugar → 3:2

Ratio

Write the definition:



3 : 2

5 eggs vs. 3 eggs — which recipe uses more?

Comparison

Write the definition:



3 : 2

3 cups flour : 2 cups sugar (ingredient to ingredient)

Part-to-part

Write the definition:



3 : 2

3 cups flour out of 5 cups total → 3:5

Part-to-whole

Write the definition:



3 : 2

3 to 2 can be written as 3:2

Colon notation

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can write and describe a ratio that compares two quantities.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Ratio

Comparison

Part-to-part

Part-to-whole

Colon notation

1

A comparison of two quantities, like 3 to 5, is a _____.

2

Showing how two amounts relate to each other is a _____.

3

A ratio that compares one part of a group to another part is a _____ ratio.

4

A ratio that compares one part of a group to the total is a _____ ratio.

5

Writing a ratio with a colon, like 3:5, is using _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: A recipe uses 4 cups of milk and 1 cup of cream. What is the ratio of milk to cream?

- A. 4:1
- B. 1:4
- C. 4:5
- D. 5:1

Step 1 A ratio compares quantities in the order given.

Step 2 Milk to cream = 4 to 1, or 4:1.

 **Answer:** A. 4:1

 We do — together

Solve this one with your class using the same steps.

Problem: Chef Reyes uses 6 strawberries and 2 bananas in a smoothie. What is the part-to-whole ratio of bananas to total fruit?

- A. 2:8
- B. 6:2
- C. 2:6
- D. 8:2

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: In a class of 30 students, 18 are girls. What is the part-to-part ratio of boys to girls?

- A. 12:18
- B. 18:12
- C. 18:30
- D. 12:30

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Kitchen Crew: A cooking class has 5 student chefs and 7 student helpers, for 12 students in all. What is the part-to-whole ratio of student chefs to the whole class?

- A. 5:12
- B. 5:7
- C. 7:12
- D. 12:5

Show your work:

2. “For every” language: Chef Reyes says, “My glaze uses 9 spoons of honey for every 4 spoons of lemon.” Written in colon notation, the ratio of honey to lemon is:

- A. 9:4
- B. 4:9
- C. 9:13
- D. 13:4

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A bag of marbles has 8 red, 6 blue, and 4 green marbles. Write three different ratios using these quantities: one part-to-part ratio, one part-to-whole ratio, and explain the difference.

Show your work:

Reflect — Exit Ticket

A bag of trail mix contains 8 peanuts and 5 raisins. What is the ratio of raisins to total pieces?

- A. 5:13
- B. 8:5
- C. 5:8
- D. 13:5

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Ratio
2. **Notes 2:** Comparison
3. **Notes 3:** Part-to-part
4. **Notes 4:** Part-to-whole
5. **Notes 5:** Colon notation
6. **We Try (notes):** A. 2:8 — *Total fruit = $6 + 2 = 8$. The part-to-whole ratio of bananas to total fruit is 2:8.*
7. **You Try (notes):** A. 12:18 — *Boys = $30 - 18 = 12$. Part-to-part ratio of boys to girls = 12:18. (Order matters!)*
8. **Try It 1:** A. 5:12 — *The whole class is $5 + 7 = 12$ students. Part-to-whole compares one part to the total, so chefs to whole class is 5:12.*
9. **Try It 2:** A. 9:4 — *“For every 9 honey to 4 lemon” is a part-to-part comparison. In colon notation, honey to lemon is 9:4.*
10. **Exit Ticket:** A. 5:13 — *Total pieces = $8 + 5 = 13$. The ratio of raisins to total is 5:13.*